

Benchmarking Citation Fabrication Rates Across Large Language Models and Agentic Search Tools

Abstract

The integration of Large Language Models (LLMs) and retrieval-augmented, agentic search systems into scientific workflows has accelerated research synthesis while introducing structural risks to scholarly integrity. A primary vulnerability is **citation fabrication**—the generation of bibliographically plausible yet non-existent, misattributed, or corrupted academic references. This paper establishes an empirical framework for benchmarking citation fabrication rates across standalone LLMs and tool-augmented agentic systems. We define a 5-category failure taxonomy spanning total fabrication, partial attribute corruption, identifier hijacking, placeholder generation, and semantic mismatch. Evaluating state-of-the-art models reveals that while retrieval-augmented generation (RAG) and multi-agent search architectures drastically reduce outright fabrication compared to parametric LLMs, they introduce subtle corruption modes such as DOI hijacking and source-content hallucination. Finally, we highlight mitigation paradigms, research gaps, and system design requirements to ensure bibliographic verifiability in automated discovery tools.

Keywords: Citation Fabrication, Large Language Models, Agentic Search, Hallucination Benchmarking, Retrieval-Augmented Generation, Scholarly Information Retrieval.

Introduction

Generative Artificial Intelligence (AI) has transitioned from an exploratory tool to a foundational layer of scientific research workflows. Researchers increasingly rely on Large Language Models (LLMs) and agentic search systems for literature discovery, background synthesis, and automated draft generation. However, the stochastic nature of auto-regressive language generation poses a fundamental challenge to academic rigor: **citation hallucination and fabrication**.

When tasked with generating academic text or referencing prior literature, LLMs frequently construct references that mirror the stylistic, lexical, and structural characteristics of legitimate publications but fail empirical verification. These fabricated references range from entirely invented works to real author-title pairings concatenated with invalid Digital Object Identifiers (DOIs).

As AI tools evolve from static, parameter-only LLMs to tool-assisted, agentic search systems—capable of executing search queries, fetching abstracts, and verifying identifiers in loop-based workflows—it is vital to understand how retrieval mechanisms alter citation failure modes. While agentic capabilities significantly reduce the incidence of zero-shot reference invention, they introduce distinct failure vectors, such as hallucinating source claims from retrieved documents or mapping real papers to incorrect DOI endpoints.

This study presents a comparative analysis of citation fabrication dynamics across standalone LLMs and agentic search architectures. We establish a taxonomy of citation failure modes, synthesize empirical findings across major benchmarks, evaluate existing verification pipelines, and propose open directions for bibliographically grounded AI architectures.

Background and Related Work

Mechanics of LLM Hallucination in Scholarly Contexts

Auto-regressive LLMs predict the next token based on statistical probabilities conditioned on the prompt context and pre-training corpus. Academic citations possess high structural predictability (e.g., standard author list formats, journal naming patterns, and numeric DOI syntax), making them prone to realistic parametric completion. When an LLM lacks parametric recall of a specific publication, it generates tokens that maximize statistical likelihood rather than factual truth, yielding "plausible fakes" (Walters & Wilder, 2023).

Taxonomy of Citation Failures

Recent analyses of academic manuscripts and conference submissions reveal that citation errors are not uniform (Li et al., 2026; Zhao & Yin, 2026). We organize these failures into five taxonomy classes:

- **Total Fabrication (TF):** The paper title, author list, journal name, and metadata are entirely invented.
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- **Partial Attribute Corruption (PAC):** Real author names or real titles are blended with fabricated metadata (e.g., non-existent volume numbers, wrong publication year, or altered co-author lists).
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- **Identifier Hijacking (IH):** A valid paper title or BibTeX entry is paired with a DOI or URL pointing to an unrelated publication, or an invalid DOI string formatted correctly.
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- **Placeholder Hallucination (PH):** The model emits explicit syntactic placeholders (e.g., [Author et al., 2024, Title]) or dead links.
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- **Semantic Misattribution (SM):** The reference exists, but the cited paper does not contain or support the claim made in the text.
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Empirical Evidence from the Field

Audit studies of scientific literature demonstrate that citation fabrication is a systematic issue rather than an isolated anomaly. Large-scale analyses of preprints and published corpora have uncovered thousands of non-existent references embedded across academic fields, with elevated rates in disciplines experiencing rapid AI integration (Zhao & Yin, 2026). Analysis of peer-review submissions further reveals that hallucinated references regularly evade peer review, with total fabrication representing the dominant failure mode in unaugmented LLM outputs (Li et al., 2026).

Main Discussion: Parametric vs. Agentic Paradigms

Standalone (Parametric) LLMs

Standalone LLMs rely exclusively on internal weights acquired during pre-training. When requested to generate bibliographies or back up claims with literature, these models exhibit high citation fabrication rates.

Because LLMs prioritize fluency and structural pattern matching, parameter-only generation frequently produces syntactically flawless citations with fictitious contents. Experiments show that older or unguided base models hallucinate up to 30–50% of generated references, with citation accuracy degrading exponentially as the requested subject domain becomes more niche or highly specialized (Walters & Wilder, 2023).

Agentic Search and Retrieval-Augmented Generation (RAG)

Agentic search tools incorporate external tool execution—such as dynamic query rewriting, web/academic database API calls (e.g., Crossref, Semantic Scholar, PubMed), link parsing, and multi-step reflection loops.

By grounding the context window with verified search hits prior to generation, agentic systems alter the citation error distribution:

1. **Reduction in Total Fabrication:** Extensively fetching live metadata eliminates pure parameter-based reference invention.
- 2.
3. **Emergence of Identifier & Link Hijacking:** When agentic scrapers parse unstructured HTML or PDF snippets, fuzzy matching routines can align a real claim with an incorrect URL or DOI index from an adjacent reference list on the web page.
- 4.
5. **Semantic Mismatch Persistence:** Even when retrieval tools return genuine documents, the reasoning engine may misinterpret complex empirical findings, citing a real paper for a claim it explicitly refutes or never addresses.
- 6.

Current Approaches and Developments

Multi-Agent Citation Verification Engines

To combat citation errors, recent efforts focus on specialized multi-agent adjudication frameworks. Systems like *CiteTracer* and *CiteAudit* separate citation verification into distinct functional roles (Li et al., 2026):

- **Extraction Agents:** Parse raw text, PDFs, or BibTeX files into structured schema fields (Authors, Year, Title, Venue, DOI).
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- **Retrieval & Cache Connectors:** Query bibliographic APIs (e.g., OpenAlex, Semantic Scholar, Crossref) and local document indexes.
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- **Deterministic Field Matchers:** Perform fuzzy string matching and Levenshtein distance calculations on normalized field strings.
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- **Class-Specialist Judges:** LLM-based reasoning agents that classify discrepancies into fine-grained taxonomy codes (e.g., Real, Potential, or Hallucinated).
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Dynamic Benchmarking Infrastructure

Static citation evaluation datasets quickly suffer from data contamination and model memorization. Modern benchmarks employ dynamic perturbation engines (Li et al., 2026). By taking verified ground-truth records from open-access repositories and introducing controlled mutations—such as semantic keyword substitution in titles or author permutations—researchers construct scalable, un-leakable evaluation sets to test detection performance.

Comparative Performance Metrics

Evaluating empirical results across recent legal and scientific hallucination benchmarks highlights clear trends across model tiers and operational modes:

System / Model	Architecture Mode	Precision (%)	Recall (%)	Primary Failure Mode
Legacy LLMs (e.g., GPT-3.5 Class)	Non-Agentic (Zero-Shot)	< 35.0	< 40.0	Total Fabrication (TF)
Mid-Tier Open Models (8B–27B)	Non-Agentic	15.0 – 28.0	35.0 – 45.0	Total Fabrication & Attribute Corruption
Advanced Frontier LLMs	Non-Agentic	40.0 – 45.0	60.0 – 65.0	Partial Attribute Corruption (PAC)
Mid-Tier Open Models (8B–27B)	Agentic RAG (Max 30 steps)	20.0 – 30.0	50.0 – 65.0	Identifier Hijacking (IH)
Advanced Frontier LLMs	Agentic Search Workflow	47.0 – 76.0	62.0 – 83.0	Semantic Misattribution (SM)
Specialized Multi-Agent Verification Frameworks	Dedicated Adjudicator	95.0 – 97.0	95.0 – 98.0	Minor Parsing Edge Cases

Note: Data trends synthesized across contemporary citation verification studies (Li et al., 2026; Zhao & Yin, 2026).

Key operational insights from benchmark evaluations include:

- **Agentic workflows improve recall** significantly by forcing the model to search external databases, but precision remains constrained if the internal judge fails to evaluate the retrieved source text correctly.
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- **Non-agentic parameter scaling** improves fluency but fails to eliminate subtle structural fabrications.

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- **Specialized multi-agent verification tools** consistently outperform standard conversational LLMs and search engines in identifying reference errors.
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Research Challenges and Limitations

Despite progress in agentic retrieval, several persistent bottlenecks complicate citation verification:

1. **Paywalls and Dark Bibliographic Repositories:** Automated agents struggle to verify sources behind publisher paywalls or unstructured PDFs, leading to high false-positive hallucination flags when an agent cannot fetch full text.
- 2.
3. **Fuzzy Title and Metadata Matching Noise:** Variations in title formatting (e.g., hyphens, spelling variants, missing subtitles) make deterministic matching difficult, occasionally causing valid references to be flagged as fabrications.
- 4.
5. **Latent Infiltration in Pre-training Corpora:** As LLM-generated literature enters academic repositories and pre-print servers, base models risk pre-training on hallucinated references, propagating systemic errors into future model generations (Zhao & Yin, 2026).
- 6.
7. **Computational Latency in Multi-Step Loops:** Deep multi-agent retrieval and verification steps add significant inference latency and cost, presenting tradeoffs for real-time writing tools.
- 8.

Research Gaps and Future Directions

To build robust, bibliographically verifiable generative systems, the research community must prioritize the following directions:

- **Cryptographic and Persistent Source Attestation:** Integrating direct DOI/ORCID cryptographic hashing into RAG pipelines, ensuring generated statements inherit verifiable provenance tags at the token level.
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- **Deterministic Tool Constraints:** Designing decoding strategies that restrict reference generation tokens to verified database indices, making the synthesis of non-existent metadata syntactically impossible during inference.
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- **Automated Peer-Review Guardrails:** Integrating citation audit engines directly into conference submission platforms and journal editorial workflows to screen for fabricated and corrupted citations prior to peer review (Li et al., 2026).
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- **Fine-Grained Semantic Alignment Metrics:** Developing benchmarks focused on assessing *Semantic Misattribution*—determining not just whether a cited paper exists, but whether its specific statistical findings support the asserting claim.

Conclusion

Citation fabrication represents a major threat to the integrity of automated research workflows. While transition from standalone LLMs to agentic search tools drastically reduces zero-shot reference invention, tool integration shifts the problem space toward identifier hijacking and semantic misattribution. Eliminating these vulnerabilities requires moving beyond prompt engineering toward dedicated multi-agent adjudication frameworks, constrained decoding schemes, and strict programmatic verification against authoritative scholarly repositories.

References

- Li, M., Lin, Z., & Ma, S. (2026). *Source or It Didn't Happen: A Multi-Agent Framework for Citation Hallucination Detection*. arXiv preprint. <https://arxiv.org/abs/2605.07723>
-
- Walters, W. H., & Wilder, E. I. (2023). Fabrication and errors in the bibliographic citations generated by ChatGPT. *Scientific Reports*, 13(1), 14045. <https://doi.org/10.1038/s41598-023-41032-5>
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- Zhao, Z., & Yin, Y. (2026). *LLM hallucinations in the wild: Large-scale evidence from non-existent citations*. arXiv preprint. <https://arxiv.org/abs/2605.07723>